Table 2.3 ICMR recommended dietary allowances of an expectant mother 2010

Nutrient	Normal adult women	Pregnant women
Energy Kcals		
Sedentary work	1900	+350
Moderate work	2230	+350
Heavy work	2850	+350
Protein gm	55	82.2
Fats gm	30	30
Calcium mg	600	1200
Iron mg	21	35
Vitamin A		
Retinol µg or	600	800
Carotene µg	4800	6400
Thiamine mg		
Sedentary work	1.0	
Moderate work	1.1	+0.2
Heavy work	1.4	
Riboflavin mg		
Sedentary work	1.1	
Moderate work	1.3	+0.3
Heavy work	1.7	
Niacin equivalent mg		
Sedentary work	12	
Moderate work	14	+2
Heavy work	16	
Pyridoxine mg	2.0	2.5
Ascorbic acid mg	40	60
Dietary folate µg	200	500
Vitamin B12 µg	1.0	1.2
Magnesium mg	310	310
Zinc	10	12

c) Minerals

i) Calcium

Calcium requirement suggested by ICMR for an adult woman is 600mg / day. Requirement increases during pregnancy to 1200 mg/ day. Increased intake of calcium by mother is highly essential, not only for the classification of foetal bones and teeth but also for protection of calcium resources of mother to meet the high demands during lactation. Dairy products are a primary source of calcium. Green leafy vegetables like agathi and gingelly seeds also contribute to calcium.

ii) Iron

Normal iron requirement for an adult woman is 21 mg/ day, ICMR requirements during pregnancy is 35 mg / day. Additional iron requirement during pregnancy is computed from iron needs for foetal growth (250 mg), expansion of maternal tissue including the red cell mass during pregnancy, (400mg) the iron content of placenta and the blood loss during parturition (250mg). There is, however, saving (150 mg) due to cessation of menstruation (amenorrhea). The iron absorption is better when taken with Vitamin C rich fruits i.e. fruits like 'amla' (Indian gooseberry), guava and other citrus fruits. Liver, dried beans, dried fruits, green leafy vegetables, eggs, enrich serials and iron fortified salt provide additional sources of iron.

d) Vitamins

i) Vitamin A

Normal Requirements of ß - carotene for an adult woman is 4800 µg and during pregnancy it is increased to 6400 µg. Same level is suggested during pregnancy. Liver, egg yolk, butter, dark green and yellow vegetables and fruits are good sources of vitamin A.

ii) Vitamin D

Vitamin D is a highly essential as it enhances the maternal calcium absorption. Its active form calcidiol and calcitriol cross the placenta with ease and play an important role in calcium metabolism of the foetus. Maternal deficiency of vitamin D results in neonatal hypocalcaemia and hypoplasia.

iii) Thiamine, Riboflavin, Niacin, Pyridoxine

Since the requirements of B complex vitamins like thiamine, riboflavin, niacin

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